Homework # 2.

1. Let $\mathcal{J}$ be a collection of non-degenerate closed intervals on $\mathbb{R}$, and let A be the set of centers of all intervals in $\mathcal{J}$. Assume that for every $\varepsilon > 0$ and every center $x \in A$ there exists an interval $I \in \mathcal{J}$ centered at x with $|I| < \varepsilon$. Finally, assume that

$$D = \sup[|I| : \ I \in \mathcal{J}] < +\infty.$$

Show that there exist two sub-collections $\mathcal{F}_1$ and $\mathcal{F}_2$ of intervals in $\mathcal{J}$ so that each $\mathcal{F}_{1,2}$ is a countable collection of pairwise disjoint intervals and the set A is covered by $\mathcal{F}_1 \cup \mathcal{F}_2$.

2. Let μ be a Borel measure on $\mathbb{R}$ and $\mathcal{F}$ any collection of non-degenerate closed intervals. Let A be the set of centers of the intervals in $\mathcal{F}$. Assume that $\mu(A) < +\infty$ and that for every $\varepsilon > 0$ and every center $x \in A$ there exists an interval $I \in \mathcal{J}$ centered at x with $|I| < \varepsilon$. Then for every open set $U \subset \mathbb{R}$ there exists a pairwise disjoint countable sub-collection $\mathcal{J}$ of intervals in $\mathcal{F}$ so that

$$\bigcup_{I \in \mathcal{J}} I \subseteq U,$$

and

$$\mu\Big(A \bigcap U \setminus \Big(\bigcup_{I \in \mathcal{J}} I\Big)\Big) = 0.$$

3. Construct a monotone function that is discontinuous on a dense set on $[0, 1]$.

4. Let E_k be a sequence of measurable sets such that

$$\sum_{k=1}^{\infty} \mu(E_k) < +\infty.$$

Show that then almost all x lie in at most finitely many of the sets E_k.

5. Let ϕ be a non-negative continuous function on $\mathbb{R}^n$ such that $\int \phi = 1$. Given $t > 0$ define $\phi_t(x) = t^{-n}\phi(x/t)$. Show that if $g \in C^\infty(\mathbb{R}^n)$ with compact support then

$$\phi_t(g) = \int_{\mathbb{R}^n} \phi_t(x) g(x) dx \to g(0).$$

Because of that ϕ_t is called an approximation of identity. How much can you weaken the regularity assumptions on ϕ and g?